



End-to-End Database & Data Warehouse Design Project

From OLTP to OLAP — A Complete Data Architecture Guide

9 Phases • Complete Documentation • SQL
Scripts • ETL Design



Project Overview

This project focuses on designing a complete data system — starting from a Transactional Database (OLTP) and progressing through to a full Data Warehouse (OLAP). The goal is to build a deep understanding of how modern data architectures are structured and how information flows through each layer.

Core Learning Objectives

- Understand the difference between operational (OLTP) and analytical (OLAP) systems
- Design and implement a normalized transactional database
- Build an efficient ETL pipeline to move data between systems
- Architect a scalable Data Warehouse using dimensional modeling
- Write analytical queries and generate business insights

Project Phases at a Glance

Phase	Title	Key Output
1	Business Understanding	KPIs, Business Questions
2	OLTP Database Design	ERD, Normalized Schema
3	OLTP Workload Simulation	Sample Data, Transactions
4	OLTP Limitations Analysis	Performance Report
5	ETL Design	Flow Diagram, Mapping Doc
6	Data Warehouse Design	Star/Snowflake Schema
7	Advanced Modeling Concepts	SCD, Surrogate Keys
8	Performance & Scalability	Indexing & Partitioning Plan
9	Analytical Queries & Reporting	SQL Queries, Results



Phase 1: Business Understanding

Define the business context and analytical goals

Objective

Select a complex business domain and deeply understand its operational and analytical needs before any technical design begins. Recommended domains include:

- Telecom System
- Retail (Stores + Online)
- Banking System
- Healthcare System

Tasks

- Describe the business scenario in detail
- Define the main business problem to be solved
- Identify key stakeholders (e.g., managers, analysts, executives)
- Define a minimum of 5 meaningful KPIs

Example KPIs

KPI	Description
Revenue	Total income generated over a period
Customer Retention	% of customers retained month-over-month
Daily Transactions	Volume of transactions processed per day
Product Performance	Sales and margin by product/SKU
Avg. Transaction Value	Mean value per completed transaction

Deliverables

- Business overview document or presentation
- Prioritized list of business questions the system must answer

Guiding Questions

Who will use this system, and what decisions will they make with it?



What does success look like for each stakeholder group?

Phase 2: OLTP Database Design

Design a fully normalized transactional database

Objective

Design a transactional database system that accurately models real-world business operations, with proper normalization and integrity constraints.

Requirements

- Minimum 7–10 tables in the schema
- Include One-to-Many relationships
- Include Many-to-Many relationships using junction/bridge tables
- Apply normalization up to 3NF or BCNF

Constraints to Define

- Primary Keys on every table
- Foreign Keys enforcing referential integrity
- Unique Constraints on business identifiers
- NOT NULL Constraints on required fields
- Business rule constraints (e.g., balance cannot be negative, quantity must be > 0)

Deliverables

- Entity Relationship Diagram (ERD)
- DDL SQL scripts for all tables
- Documentation of business rules embedded in the schema

Guiding Questions

What are the consequences of incorrectly designed relationships?

How would you detect and resolve anomalies in your design?



Phase 3: OLTP Workload Simulation

Populate and stress-test the transactional database

Objective

Simulate real-world database usage by inserting representative data and performing typical transactional operations to validate your schema design.

Tasks

- Insert a meaningful volume of sample data across all tables
- Perform INSERT operations for new business events
- Perform UPDATE operations to reflect state changes
- Perform DELETE operations with cascading or soft-delete logic
- Execute concurrent transactions to test constraint enforcement

Deliverables

- SQL seed scripts with sample data
- Transaction scripts demonstrating CRUD operations

Guiding Questions

Can your system handle continuous, concurrent operations efficiently?

Do your constraints correctly prevent invalid data from being inserted?

Phase 4: OLTP Limitations Analysis

Understand why OLTP is unsuitable for analytics

Objective

Demonstrate through practical examples why transactional databases, while optimized for writes, are poorly suited for complex analytical workloads.

Tasks

- Write complex analytical queries involving multiple JOINS across 4+ tables



- Measure and document query execution times
- Explain the performance challenges observed (locking, I/O, row-by-row scanning)
- Describe structural limitations: no pre-aggregated data, no historical snapshots

Deliverables

- Technical report with annotated SQL examples
- Execution plan screenshots (EXPLAIN output)
- Summary of OLTP vs OLAP suitability for analytics

Guiding Questions

Why are OLTP systems not ideal for reporting and analytics?

What would happen to production performance if analysts ran heavy queries directly?



Phase 5: ETL Design

Build the data pipeline from OLTP to Data Warehouse

Objective

Design and document the Extract, Transform, Load process that moves data from the operational OLTP system into the analytical Data Warehouse.

Tasks

- Define source-to-target column mappings for every table
- Identify and handle missing values with appropriate strategies
- Detect and eliminate duplicate records
- Resolve data type inconsistencies between source and target
- Design transformation logic (e.g., currency conversion, date standardization)

Optional Tools

- SQL — stored procedures or views for transformation
- Python — pandas/SQLAlchemy for scripted ETL
- Apache NiFi or similar ETL orchestration tool

Deliverables

- ETL flow diagram showing Extract → Transform → Load stages
- Source-to-target mapping document with transformation rules

Guiding Questions

Is your source data clean enough to load directly, or does it require significant transformation?

How will you handle late-arriving or out-of-order data?

Phase 6: Data Warehouse Design

Build a dimensional model for analytical queries



Objective

Design a Data Warehouse schema optimized for fast analytical queries, using dimensional modeling techniques that separate measures from context.

Requirements

- Use Star Schema (preferred) or Snowflake Schema
- Include 1–2 Fact Tables with clearly defined grain
- Include 3–5 Dimension Tables

Measure Types to Define

Measure Type	Description	Example
Additive	Can be summed across all dimensions	Revenue, Units Sold
Semi-additive	Can be summed across some dimensions only	Account Balance, Inventory Level
Non-additive	Cannot be meaningfully summed	Ratios, Percentages, Prices

Deliverables

- Data Warehouse schema diagram (Star or Snowflake)
- Table descriptions with grain, measures, and keys defined

Guiding Questions

What does each row in your fact table represent — one transaction? One day? One product?

Which dimensions are shared across multiple fact tables?

Phase 7: Advanced Modeling Concepts

Apply enterprise-grade data warehouse patterns

Objective

Extend your Data Warehouse design with advanced concepts that handle real-world complexities like historical tracking, degenerate attributes, and sparse event data.



Required Concepts

Surrogate Keys

Replace natural business keys with system-generated integer keys for all dimension tables to ensure stability and join performance.

Slowly Changing Dimensions (SCD)

- Type 1 — Overwrite: Update the existing record. No history preserved.
- Type 2 — Add New Row: Insert a new record with effective dates. Full history maintained.

Optional Advanced Concepts

- Junk Dimension — Combine low-cardinality flags into a single dimension table
- Degenerate Dimension — Store transaction IDs directly in the fact table (no separate dim table)
- Factless Fact Table — Record events that have no measurable values (e.g., enrollment, attendance)

Deliverables

- Written explanation of each concept applied, linked to your specific design
- SCD Type 2 example showing before/after rows with effective dates

Guiding Questions

How will you track changes in customer address, category, or name over time?

Which dimensions in your design are candidates for SCD Type 2?



Phase 8: Performance & Scalability

Design for large-scale data volumes

Objective

Ensure your Data Warehouse can handle millions of records efficiently through a deliberate indexing and partitioning strategy.

Tasks

Indexing Strategy

- Define bitmap indexes on low-cardinality dimension columns
- Define B-tree indexes on foreign keys and frequently filtered columns
- Document which columns are covered by each index and why

Partitioning Strategy

- Apply range partitioning on fact table date columns
- Consider hash partitioning for even data distribution
- Explain partition pruning and how it reduces query scan range

Deliverables

- Indexing plan document with justification for each index
- Partitioning design with expected query performance improvement

Guiding Questions

Can your system handle hundreds of millions of fact table rows without degradation?

How does partition pruning reduce I/O in date-range analytical queries?

Phase 9: Analytical Queries & Reporting

Extract business insights from the Data Warehouse

Objective



Demonstrate the value of the Data Warehouse by writing analytical SQL queries that answer key business questions and showcase the performance advantages of dimensional modeling.

Required Query Types

- Top N customers by revenue or transaction count
- Monthly and yearly revenue trends with period-over-period comparison
- Best-performing products by category and region
- Customer retention and churn rate analysis
- Rolling averages and running totals using window functions

Deliverables

- Well-commented SQL scripts for each analytical query
- Result screenshots or exported data tables
- Brief narrative interpreting the business insight from each query

Guiding Questions

Are analytical queries significantly faster on the Data Warehouse vs the OLTP system? Why?

How do window functions enable calculations that would be complex on OLTP?



Final Deliverables Summary

Deliverable	Format	Phase
Full project documentation	PDF / Word Document	All
ERD (OLTP)	Diagram (draw.io / Visio)	Phase 2
Data Warehouse Schema	Diagram (Star/Snowflake)	Phase 6
All SQL scripts	SQL files (.sql)	Phases 2–4, 9
ETL design document	Diagram + Mapping table	Phase 5
Advanced modeling explanations	Written in documentation	Phase 7
Performance strategy	Written in documentation	Phase 8
Analytical query results	Screenshots / CSV exports	Phase 9
Final presentation	PowerPoint / Slides	All

Bonus (Optional Enhancements)

- Conduct a formal performance comparison between OLTP and Data Warehouse query execution
- Build an interactive dashboard in Power BI or Tableau connected to the Data Warehouse
- Use a real publicly available dataset instead of synthetic data (e.g., Kaggle, government open data)
- Implement incremental ETL loads instead of full refreshes
- Add a data quality layer with automated validation checks